



SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

Q1

Handling ArithmeticException

10 marks

Q2

Handling ArrayIndexOutOfBoundsException

10 marks

Q3

Handling NumberFormatException

10 marks

Q4

Handling Multiple Exceptions

10 marks

4/4 answered

Q4 Handling Multiple Exceptions

10 marks

Write a Java program that performs the following operations:

Accept two integers from the user.

Divide the first number by the second.

Store the result in an array.

Handle both ArithmeticException and ArrayIndexOutOfBoundsException using multiple catch blocks.

Your Code

```
1 import java.util.Scanner;
2 public class MultipleExceptionHandling {
3     public static void main(String[] args) {
4         Scanner scanner = new Scanner(System.in);
5         int[] results = new int[1];
6         try {
7             System.out.print("Enter first integer: ");
8             int num1 = scanner.nextInt();
9             System.out.print("Enter second integer: ");
10            int num2 = scanner.nextInt();
11            int divisionResult = num1 / num2;
12            results[0] = divisionResult;
13            System.out.println("Result: " + results[0]);
14        } catch (ArithmeticException e) {
15            System.out.println("Error: Cannot divide by zero! (" + e + ")");
16        } catch (ArrayIndexOutOfBoundsException e) {
17            System.out.println("Error: Array index is out of bounds! (" + e + ")");
18        }
```

Input

```
1
2
```

Output

```
Enter first integer: Enter second
integer: Result: 0
Program execution completed.
```

Write a Java program to accept a number as a string and convert it into an integer using `Integer.parseInt()`. If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 import java.util.Scanner;
2 public class NumberFormatExceptionExample {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter a number as string: ");
6         String input = sc.nextLine();
7         try {
8             int num = Integer.parseInt(input);
9             System.out.println("Converted integer: " + num);
10        } catch (NumberFormatException e) {
11            System.out.println("Invalid input! Please enter a valid integer.");
12        }
13        sc.close();
14    }
15 }
```

Input

1
2

Output

Enter a number as string: Converted
integer: 1

Run

Save

Q1
Handling ArithmeticException
10 marks

Q2
Handling
ArrayIndexOutOfBoundsException
10 marks

Q3
Handling
NumberFormatException
10 marks

Q4
Handling Multiple Exceptions
10 marks

4/4 answered

ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 import java.util.Scanner;  
2 public class ArrayExceptionHandler {  
3     public static void main(String[] args) {  
4         Scanner scanner = new Scanner(System.in);  
5         int[] numbers = {10, 20, 30, 40, 50};  
6         try {  
7             System.out.print("Enter an index (0-4): ");  
8             int index = scanner.nextInt();  
9             System.out.println("Element at index " + index + ": " + numbers[index]);  
10        } catch (ArrayIndexOutOfBoundsException e) {  
11            System.out.println("Error: The index you entered is outside the valid range.");  
12        } catch (Exception e) {  
13            System.out.println("Error: Please enter a valid integer.");  
14        } finally {  
15            scanner.close();  
16        }  
17    }  
18 }
```

Input

2

Output

Enter an index (0-4): Element at
index 2: 30



Search



ENG
US

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Q1Handling ArithmeticException
10 marks**Q2**Handling
ArrayOutOfBoundsException
10 marks**Q3**Handling
NumberFormatException
10 marks**Q4**Handling Multiple Exceptions
10 marks

4/4 answered

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int numerator = sc.nextInt();
6         int denominator = sc.nextInt();
7         try {
8             int result = numerator / denominator;
9             System.out.println("Result: " + result);
10        } catch (ArithmeticException e) {
11            System.out.println("Error: Division by zero is not allowed.");
12        }
13        sc.close();
14    }
15 }
```

Input10
2**Output**

Result: 5

 Run Save